Zero Divider Cpp

Component Design Document

1 Description

This component provides commands that exercise C++ fault and arithmetic behavior so that a target's response to them can be observed. The 'Raise_Exception_in_Cpp' command explicitly raises a C++ exception. When a user-implemented C++ Termination Handler (TH) is configured to forward termination events to the Ada Last Chance Handler (LCH), this command allows verification that the exception propagation pathway is functioning correctly. The 'Int_Divide_By_Zero_In_Cpp' and 'Fp_Divide_By_Zero_In_Cpp' commands are informational rather than authoritative. Each performs a division by zero in C++ and reports what the target does with the result. That outcome depends on the target architecture, the compiler, and the compiler and runtime flags the project is built with, so this component makes no claim about which outcome any particular configuration produces. Integer division by zero is undefined behavior in C++, so a target may trap or may return a value. Floating point division by zero on a target that implements IEEE 754 typically yields a signed infinity for a non-zero dividend and a NaN for a zero dividend, though whether such a value reaches Ada or raises an exception on the way is itself configuration dependent. Whenever a division returns to Ada without raising an exception, the returned value is reported in an event. Run these commands on the intended target to establish how that configuration behaves.

2 Requirements

The requirements for the Zero Divider Cpp component.

1. The component shall provide a command, that when executed, causes an unhandled exception to be thrown in C++.
2. The component shall provide a command, that when executed, performs an integer division by zero in C++ and reports the outcome.
3. The component shall provide a command, that when executed, performs a floating point division by zero in C++ and reports the outcome.
4. The component shall provide a protection mechanism that protects these commands from being executed accidentally.

3 Design

3.1 At a Glance

Below is a list of useful parameters and statistics that give a quick look into the makeup of the component.

- **Execution** - *passive*
- **Number of Connectors** - 4
- **Number of Invokee Connectors** - 1

- **Number of Invoker Connectors** - 3
- **Number of Generic Connectors** - *None*
- **Number of Generic Types** - *None*
- **Number of Unconstrained Arrayed Connectors** - *None*
- **Number of Commands** - 3
- **Number of Parameters** - *None*
- **Number of Events** - 8
- **Number of Faults** - *None*
- **Number of Data Products** - *None*
- **Number of Data Dependencies** - *None*
- **Number of Packets** - 1

3.2 Diagram

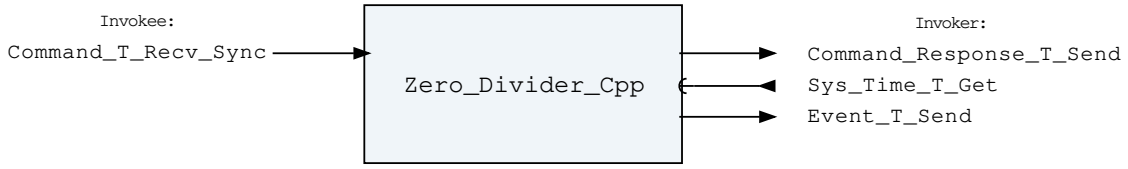

Figure 1: Zero Divider Cpp component diagram.

3.3 Connectors

Below are tables listing the component's connectors.

3.3.1 Invokee Connectors

The following is a list of the component's *invokee* connectors:

Table 1: Zero Divider Cpp Invokee Connectors

Name	Kind	Type	Return_Type	Count
Command_T_Recv_Sync	recv_sync	Command.T	-	1

Connector Descriptions:
- **Command_T_Recv_Sync** - The command receive connector

3.3.2 Invoker Connectors

The following is a list of the component's *invoker* connectors:

Table 2: Zero Divider Cpp Invoker Connectors

Name	Kind	Type	Return_Type	Count
Command_Response_T_Send	send	Command_Response.T	-	1
Sys_Time_T_Get	get	-	Sys_Time.T	1

Event_T_Send	send	Event.T	-	1

Connector Descriptions:

- **Command_Response_T_Send** - This connector is used to register and respond to the component's commands.
- **Sys_Time_T_Get** - The system time is retrieved via this connector.
- **Event_T_Send** - Events are sent out of this connector.

3.4 Interrupts

This component contains no interrupts.

3.5 Initialization

Below are details on how the component should be initialized in an assembly.

3.5.1 Component Instantiation

This component contains no instantiation parameters in its discriminant.

3.5.2 Component Base Initialization

This component contains no base class initialization, meaning there is no `init_Base` subprogram for this component.

3.5.3 Component Set ID Bases

This component contains commands, events, packets, faults, or data products that require a base identifier to be set at initialization. The `set_Id_Bases` procedure must be called with the following parameters:

Table 3: Zero Divider Cpp Set Id Bases Parameters

Name	Type
Command_Id_Base	Command_Types.Command_Id_Base
Event_Id_Base	Event_Types.Event_Id_Base
Packet_Id_Base	Packet_Types.Packet_Id_Base

Parameter Descriptions:

- **Command_Id_Base** - The value at which the component's command identifiers begin.
- **Event_Id_Base** - The value at which the component's event identifiers begin.
- **Packet_Id_Base** - The value at which the component's unresolved packet identifiers begin.

3.5.4 Component Map Data Dependencies

This component contains no data dependencies.

3.5.5 Component Implementation Initialization

The calling of this implementation class initialization procedure is mandatory. The magic number and the time to sleep before each command executes are provided at instantiation. The `init` subprogram requires the following parameters:

Table 4: Zero Divider Cpp Implementation Initialization Parameters

Name	Type	Default Value
Magic_Number	Magic_Number_Type	*None provided*
Sleep_Before_Execute_Ms	Natural	1_000

Parameter Descriptions:

- **Magic_Number** - As commands to this component crash the system, provide this number as the key value of the safety interlock mechanism that guards against unintentional execution of commands in this component.
- **Sleep_Before_Execute_Ms** - The number of milliseconds to sleep after receiving a command before executing its implementation. This allows time for any events to be written by the component, if desired.

3.6 Commands

Commands for the Zero Divider Cpp component.

Table 5: Zero Divider Cpp Commands

Local ID	Command Name	Argument Type
0	Int_Divide_By_Zero_In_Cpp	Int_Divide_By_Zero_In_Cpp_Arg.T
1	Fp_Divide_By_Zero_In_Cpp	Fp_Divide_By_Zero_In_Cpp_Arg.T
2	Raise_Exception_In_Cpp	Packed_Magic_Number.T

Command Descriptions:

- **Int_Divide_By_Zero_In_Cpp** - Performs an integer division by zero in C++ and reports the outcome. Integer division by zero is undefined behavior in C++, so what happens depends on the target and the flags the project is built with. A target may trap, or may return a value that is reported in an event. This command is informational, run it to establish what a given configuration does. You must provide the correct value for the magic number and an integer dividend for this command to execute.
- **Fp_Divide_By_Zero_In_Cpp** - Performs a floating point division by zero in C++ and reports the outcome. A target that implements IEEE 754 typically produces a signed infinity for a non-zero dividend and a NaN for a zero dividend, but whether such a value reaches Ada or raises an exception on the way depends on the target and the flags the project is built with. This command is informational, run it to establish what a given configuration does. You must provide the correct value for the magic number and a floating point dividend for this command to execute.
- **Raise_Exception_In_Cpp** - Raises a standard exception in C++. You must provide the correct value for the magic number argument of this command for it to be executed.

3.7 Parameters

The Zero Divider Cpp component has no parameters.

3.8 Events

Below is a list of the events for the Zero Divider Cpp component.

Table 6: Zero Divider Cpp Events

Local ID	Event Name	Parameter Type
0	Raising_Exception_In_Cpp	Packed_U32.T
1	Raise_Exception_In_Cpp_No_Exception	-
2	Int_Dividing_By_Zero_In_Cpp	Packed_U32.T
3	Int_Divide_By_Zero_No_Exception	Packed_I32.T
4	Fp_Dividing_By_Zero_In_Cpp	Packed_U32.T
5	Fp_Divide_By_Zero_No_Exception	Packed_F32.T
6	Invalid_Magic_Number	Packed_Magic_Number.T
7	Invalid_Command_Received	Invalid_Command_Info.T

Event Descriptions:

- **Raising_Exception_In_Cpp** - A Raise_Exception_In_Cpp command was received and the magic number was correct. The exception will be raised in N milliseconds, where N is provided as the event parameter.
- **Raise_Exception_In_Cpp_No_Exception** - The C++ exception raise did not propagate as expected. This event should never fire under normal operation and indicates the target does not propagate C++ exceptions to Ada as expected.
- **Int_Dividing_By_Zero_In_Cpp** - An Int_Divide_By_Zero_In_Cpp command was received and the magic number was correct. The division will occur in N milliseconds, where N is provided as the event parameter.
- **Int_Divide_By_Zero_No_Exception** - The integer divide by zero in C++ returned to Ada without raising an exception. This is one of the outcomes the command exists to discover, not an anomaly. The parameter is the raw result returned by C++.
- **Fp_Dividing_By_Zero_In_Cpp** - An Fp_Divide_By_Zero_In_Cpp command was received and the magic number was correct. The floating point division will occur in N milliseconds, where N is provided as the event parameter.
- **Fp_Divide_By_Zero_No_Exception** - The floating point divide by zero in C++ returned to Ada without raising an exception. This is one of the outcomes the command exists to discover, not an anomaly. The parameter is the raw result returned by C++.
- **Invalid_Magic_Number** - A command was received, but the magic number was incorrect. A magic number outside the range the type permits never reaches this event, it is rejected by command validation and reported as an invalid command instead.
- **Invalid_Command_Received** - A command was received with invalid parameters.

3.9 Data Products

The Zero Divider Cpp component has no data products.

3.10 Data Dependencies

The Zero Divider Cpp component has no data dependencies.

3.11 Packets

The packet listed here is not actually produced by this component, but instead should be produced by the implementation of the Last_Chance_Handler. This packet definition exists to ensure that the packet gets reflected in the documentation and ground system definitions.

Table 7: Zero Divider Cpp Packets

Local ID	Packet Name	Type
0x0000 (0)	Last_Chance_Handler_Packet	Packed_Exception_Occurrence.T

Packet Descriptions:

- **Last_Chance_Handler_Packet** - This packet contains information regarding an exception occurrence that triggers the Last_Chance_Handler to get invoked. This packet is not produced directly by this component, and should be produced by the last chance handler implementation. This packet definition exists to ensure that the packet gets reflected in the documentation and ground system definitions. For the ground system to identify the packet, the APID assigned to this definition must be coordinated with the APID the last chance handler implementation emits.

3.12 Faults

The Zero Divider Cpp component has no faults.

4 Unit Tests

The following section describes the unit test suites written to test the component.

4.1 *Zero_Divider_Cpp_Tests* Test Suite

This is a test suite for the Zero Divider Cpp component.

Test Descriptions:

- **Test_Bad_Magic_Number** - This test makes sure the Int_Divide_By_Zero_In_Cpp, Fp_Divide_By_Zero_In_Cpp, and Raise_Exception_In_Cpp commands do not execute if an incorrect but representable magic number is provided.
- **Test_Out_Of_Range_Magic_Number** - This test makes sure that each of the Int_Divide_By_Zero_In_Cpp, Fp_Divide_By_Zero_In_Cpp, and Raise_Exception_In_Cpp commands is rejected when it carries a magic number of 0 or 1, the two values the magic number type excludes. Such a command is caught by command validation and reported as an invalid command, so it never reaches a command handler and never reports an invalid magic number.
- **Test_Int_Divide_By_Zero_In_Cpp** - This test records how the integer division by zero behaves in the configuration the unit tests are built and run in, which is the Linux_Test target on x86-64 with GNAT numeric overflow checking (-gnato), assertions (-gnata) and full validity checking (-gnatVa) enabled. In that configuration the processor traps on the division and the GNAT Linux runtime signal manager raises a Constraint_Error, so the command never returns to report a value. The assertion below characterizes that configuration only. Another target or flag set may return a value instead, which the command reports in an event.
- **Test_Fp_Divide_By_Zero_In_Cpp** - This test records how the floating point division by zero behaves in the configuration the unit tests are built and run in, which is the Linux_Test target on x86-64 with GNAT numeric overflow checking (-gnato), assertions (-gnata) and full validity checking (-gnatVa) enabled. In that configuration C++ returns an infinity and the validity check on the Ada Short_Float result rejects it as invalid data, raising a Constraint_Error, so the command never returns to report a value. The assertion below characterizes that configuration only. Another target or flag set may deliver the infinity to Ada intact, which the command reports in an event.
- **Test_Raise_Exception_In_Cpp** - This test makes sure a C++ exception is raised and propagated.

- **Test_Invalid_Command** - This test makes sure an invalid command is rejected.

5 Appendix

5.1 Preamble

This component contains the following preamble code. This is inline Ada code included in the component model that is usually used to define types or instantiate generic packages used by the component. Preamble code is inserted as the top line of the component base package specification.

```
1 subtype Magic_Number_Type is Packed_Magic_Number.Magic_Number_Type;
```

5.2 Packed Types

The following section outlines any complex data types used in the component in alphabetical order. This includes packed records and packed arrays that might be used as connector types, command arguments, event parameters, etc..

Command.T:

Generic command packet for holding arbitrary commands

Table 8: Command Packed Record : 2080 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Command_ Header.T	-	40	0	39	-
Arg_Buffer	Command_ Types. Command_Arg_ Buffer_Type	-	2040	40	2079	Header.Arg_ Buffer_Length

Field Descriptions:

- **Header** - The command header
- **Arg_Buffer** - A buffer that contains the command arguments

Command_Header.T:

Generic command header for holding arbitrary commands

Table 9: Command_Header Packed Record : 40 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Source_Id	Command_Types. Command_Source_Id	0 to 65535	16	0	15
Id	Command_Types. Command_Id	0 to 65535	16	16	31
Arg_Buffer_Length	Command_Types. Command_Arg_Buffer_ Length_Type	0 to 255	8	32	39

Field Descriptions:

- **Source_Id** - The source ID. An ID assigned to a command sending component.
- **Id** - The command identifier
- **Arg_Buffer_Length** - The number of bytes used in the command argument buffer

Command_Response.T:

Record for holding command response data.

Table 10: Command_Response Packed Record : 56 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Source_Id	Command_Types.Command_Source_Id	0 to 65535	16	0	15
Registration_Id	Command_Types.Command_Registration_Id	0 to 65535	16	16	31
Command_Id	Command_Types.Command_Id	0 to 65535	16	32	47
Status	Command_Enums.Command_Response_Status.E	0 => Success 1 => Failure 2 => Id_Error 3 => Validation_Error 4 => Length_Error 5 => Dropped 6 => Register 7 => Register_Source	8	48	55

Field Descriptions:

- **Source_Id** - The source ID. An ID assigned to a command sending component.
- **Registration_Id** - The registration ID. An ID assigned to each registered component at initialization.
- **Command_Id** - The command ID for the command response.
- **Status** - The command execution status.

Event.T:

Generic event packet for holding arbitrary events

Table 11: Event Packed Record : 328 bits *(maximum)*

Name	Type	Range	Size (Bits)	Start Bit	End Bit	Variable Length
Header	Event_Header.T	-	72	0	71	-
Param_Buffer	Event_Types.Parameter_Buffer_Type	-	256	72	327	Header.Param_Buffer_Length

Field Descriptions:

- **Header** - The event header
- **Param_Buffer** - A buffer that contains the event parameters

Event_Header.T:

Generic event packet for holding arbitrary events

Table 12: Event_Header Packed Record : 72 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Time	Sys_Time.T	-	48	0	47
Id	Event_Types.Event_Id	0 to 65535	16	48	63
Param_Buffer_Length	Event_Types.Parameter_Buffer_Length_Type	0 to 32	8	64	71

Field Descriptions:

- **Time** - The timestamp for the event.
- **Id** - The event identifier
- **Param_Buffer_Length** - The number of bytes used in the param buffer

Fp_Divide_By_Zero_In_Cpp_Arg.T:

Argument type for the Fp_Divide_By_Zero_In_Cpp command carrying both the magic number and the floating point dividend.

Table 13: Fp_Divide_By_Zero_In_Cpp_Arg Packed Record : 64 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Magic_Number	Packed_Magic_Number.T	-	32	0	31
Dividend	Short_Float	-3.40282e+38 to 3.40282e+38	32	32	63

Field Descriptions:

- **Magic_Number** - The magic number that must match for the command to execute.
- **Dividend** - The floating point value to be divided by zero.

Int_Divide_By_Zero_In_Cpp_Arg.T:

Argument type for the Int_Divide_By_Zero_In_Cpp command carrying both the magic number and the integer dividend.

Table 14: Int_Divide_By_Zero_In_Cpp_Arg Packed Record : 64 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Magic_Number	Packed_Magic_Number.T	-	32	0	31
Dividend	Interfaces.Integer_32	-2147483648 to 2147483647	32	32	63

Field Descriptions:

- **Magic_Number** - The magic number that must match for the command to execute.
- **Dividend** - The integer value to be divided by zero.

Invalid_Command_Info.T:

Record for holding information about an invalid command

Table 15: Invalid_Command_Info Packed Record : 112 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Id	Command_Types.Command_Id	0 to 65535	16	0	15
Errant_Field_Number	Interfaces.Unsigned_32	0 to 4294967295	32	16	47
Errant_Field	Basic_Types.Poly_Type	-	64	48	111

Field Descriptions:

- **Id** - The command Id received.
- **Errant_Field_Number** - The field that was invalid. 1 is the first field, 0 means unknown field, 2**32 means that the length field of the command was invalid.
- **Errant_Field** - A polymorphic type containing the bad field data, or length when Errant_Field_Number is 2**32.

Packed_Address.T:

A packed system address.

Table 16: Packed_Address Packed Record : 64 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Address	System.Address	-	64	0	63

Field Descriptions:

- **Address** - The starting address of the memory region.

Packed_Exception_Occurrence.T:

Packed record which holds information from an Ada Exception Occurrence type. This is the type passed into the Last Chance Handler when running a full runtime. *Preamble (inline Ada definitions):*

```
type Exception_Name_Buffer is new Basic_Types.Byte_Array (0 .. 99)
  with Size => 100 * 8,
       Object_Size => 100 * 8;
type Exception_Message_Buffer is new Basic_Types.Byte_Array (0 .. 299)
  with Size => 300 * 8,
       Object_Size => 300 * 8;
```

Table 17: Packed_Exception_Occurrence Packed Record : 7712 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Exception_Name	Exception_Name_ Buffer	-	800	0	799
Exception_ Message	Exception_ Message_Buffer	-	2400	800	3199
Stack_Trace_ Depth	Interfaces. Unsigned_32	0 to 4294967295	32	3200	3231
Stack_Trace	Stack_Trace_ Addresses.T	-	4480	3232	7711

Field Descriptions:

- **Exception_Name** - The exception name.
- **Exception_Message** - The exception message.
- **Stack_Trace_Depth** - The depth of the reported stack trace.
- **Stack_Trace** - The stack trace addresses.

Packed_F32.T:

Single component record for holding packed 32-bit floating point number.

Table 18: Packed_F32 Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Short_Float	-3.40282e+38 to 3.40282e+38	32	0	31

Field Descriptions:

- **Value** - The 32-bit floating point number.

Packed_I32.T:

Single component record for holding packed signed 32-bit value.

Table 19: Packed_I32 Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Interfaces. Integer_32	-2147483648 to 2147483647	32	0	31

Field Descriptions:

- **Value** - The 32-bit signed integer.

Packed_Magic_Number.T:

A packed record which holds the magic number that guards this component's commands. The value range excludes 0 and 1 so that a command carrying either is rejected by command validation rather than reaching a command handler. *Preamble (inline Ada definitions):*

```
subtype Magic_Number_Type is Interfaces.Unsigned_32 range 2 ..
 ↪  Interfaces.Unsigned_32'Last;
```

Table 20: Packed_Magic_Number Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Magic_Number	Magic_Number_Type	2 to 4294967295	32	0	31

Field Descriptions:

- **Magic_Number** - The magic number.

Packed_U32.T:

Single component record for holding packed unsigned 32-bit value.

Table 21: Packed_U32 Packed Record : 32 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Value	Interfaces. Unsigned_32	0 to 4294967295	32	0	31

Field Descriptions:

- **Value** - The 32-bit unsigned integer.

Stack_Trace_Addresses.T:

An array of packed addresses in big endian. This is sized to easily fit a normal stack trace.

Table 22: Stack_Trace_Addresses Packed Array : 4480 bits

Type	Range	Element Size (Bits)	Length	Total Size (Bits)
Packed_Address.T	-	64	70	4480

Sys_Time.T:

A record which holds a time stamp using EMA VTC format including seconds and subseconds since epoch (1-5-1980 to 1-6-1980 midnight).

Table 23: Sys_Time Packed Record : 48 bits

Name	Type	Range	Size (Bits)	Start Bit	End Bit
Seconds	Interfaces. Unsigned_32	0 to 4294967295	32	0	31
Subseconds	Interfaces. Unsigned_16	0 to 65535	16	32	47

Field Descriptions:

- **Seconds** - The number of seconds elapsed since epoch.
- **Subseconds** - The number of 1/(2^16) sub-seconds.

5.3 Enumerations

The following section outlines any enumerations used in the component.

Command_Enums.Command_Response_Status.E:

This status enumeration provides information on the success/failure of a command through the command response connector.

Table 24: Command_Response_Status Literals:

Name	Value	Description
Success	0	Command was passed to the handler and successfully executed.
Failure	1	Command was passed to the handler not successfully executed.
Id_Error	2	Command id was not valid.
Validation_Error	3	Command parameters were not successfully validated.
Length_Error	4	Command length was not correct.
Dropped	5	Command overflowed a component queue and was dropped.
Register	6	This status is used to register a command with the command routing system.
Register_Source	7	This status is used to register command sender's source id with the command router for command response forwarding.